NEUROFUSION AI | FINAL CLINICAL DIAGNOSTIC REPORT

REPORT ID: REP-20261102-3D3404

Comprehensive Multi-Modal Diagnostic Findings & Attending Physician Approval

November 02, 2026 • 16:45 UTC

PATIENT INFORMATION	ATTENDING NEUROLOGIST	EXAMINATION DETAILS
ID: PAT-1083 **Name:** Eleanor Vance **Age/Gender:** 67 yrs / Female **Status:** Parkinson's Disease (Hoehn & Yahr Stage II)	**Name:** Dr. Clara Rodriguez, MD **License:** MD-NEURO-77491 **Dept:** Neurology & Movement Disorders **Hospital:** Metropolitan Neurological Inst.	**Scan Date:** 2026-11-02 **Report Date:** 2026-11-02 **Modality:** Voice **Device/Protocol:** NeuroFusionAI Protocol PARKINSON v2.0

PRIMARY AI PREDICTION: PARKINSON

Confidence: 89.5%
Target: Parkinson's Disease

Class Probability Distribution: Parkinson: 89.5% • Normal: 10.5%

AI Confidence Calibrated via Temperature Scaling

5. ELECTROENCEPHALOGRAPHY (EEG) NEUROPHYSIOLOGICAL ANALYSIS

Channels	Duration	Slowing Index	Dominant Band	Delta (0.5-4Hz)	Theta (4-8Hz)	Alpha (8-13Hz)	Beta (13-30Hz)
8 ch	12.5 s	1.48	**Theta**	35.0%	44.0%	14.0%	7.0%
EEG Interpretation:	Excess fronto-central theta power indicative of cortico-basal motor slowing.						

7. CLINICAL DECISION SUPPORT SYSTEM (CDSS) RECOMMENDATIONS

RISK LEVEL:
HIGH CLINICAL RISK

Urgency: Prompt Outpatient Specialist Review (Within 2-3 Weeks)

Evidence-Based Diagnostic Recommendations:
- Confirm nigrostriatal degeneration via DaTscan SPECT.
- Conduct baseline MDS-UPDRS motor examination.
- Initiate physical therapy and neurological monitoring.

HYBRID QUANTUM DECISION LAYER (HQDL) VERIFICATION

Quantum Fidelity: 0.988 **Quantum Von Neumann Entropy:** 0.114 **Quantum-Refined Confidence:** 99.4%

PATIENT LONGITUDINAL DIAGNOSTIC HISTORY

Date	Modality	Prediction	Confidence	Clinical Status
2026-11-02	Voice	Parkinson	89.5%	Verified Match
2026-11-02	Spiral	Parkinson	87.4%	Verified Match
2026-11-02	Voice	Parkinson	69.1%	Verified Match
2026-11-02	Voice	Parkinson	69.1%	Verified Match

ATTENDING PHYSICIAN OBSERVATIONS & NOTES:	CLINICAL APPROVAL STATUS
Patient presents with resting tremor and bradykinesia. MRI and EEG biomarkers corroborate nigrostriatal dysfunction. Initiate clinical motor monitoring.	Signed by: *Dr. Clara Rodriguez, MD* License ID: MD-NEURO-77491 Status: **DIGITALLY APPROVED & CERTIFIED**

REGULATORY MEDICAL DISCLAIMER & CLINICAL INTENDED USE:

NeuroFusionAI is an investigational Artificial Intelligence Clinical Decision Support System (CDSS) designed to augment neurological assessment. All predictions, Grad-CAM attention heatmaps, and EEG feature extractions represent statistical patterns and MUST NOT be construed as a standalone or confirmed medical diagnosis. Clinical treatment decisions must be made solely by a board-certified physician in conjunction with the patient's full medical history.